

Double hashing

This technique makes use of 2 hash function for finding an empty position and to insert data values.

Activity - 4

Hashing Problem

Given input { 4371, 1323, 6173, 4199, 4344, 9679, 1989 }
and a hash function $h(x) = x \pmod{10}$.

at Separate chaining hash Table

Index	Chain
0	-
1	4371
2	-
3	(1323 → 6173)
4	4344
5	-
6	-
7	-
8	-
9	4199 → 9679 → 1989

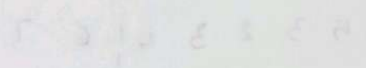
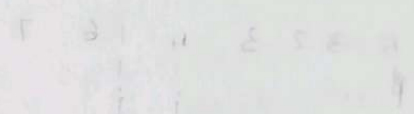
$$i=4 \quad (9+24) \bmod 10 = 3$$

$$i=5 \quad (9+36) \bmod 10 = 9$$

it can't be insert in it keep cycling through the same position.

Position	value
0	-
1	4371
2	-
3	1323
4	6173
5	9679
6	-
7	4344
8	-
9	4199

Handwritten signature



also Open Addressing -

Given:

$$h_1(x) = x \bmod 10$$

$$h_2(x) = 7 \lfloor x \bmod 7 \rfloor$$

formula:

$$h(x) = (h_1(x) + i \cdot h_2(x)) \bmod 10$$

value	h_1	$x \bmod 7$	h_2
4371	1	3	4
1323	3	0	7
6173	3	6	1
4199	9	6	1
4344	4	4	3
9679	9	5	2
1989	9	1	6

Now insert

$$4371 \rightarrow 1$$

$$1323 \rightarrow 3$$

$$6173 \rightarrow 3 \rightarrow \text{occupied} \rightarrow 3+1=4$$

$$4199 \rightarrow 9$$

$$4344 \rightarrow 4 \rightarrow \text{occupied} \rightarrow 4+3=7$$

$$9679 \rightarrow 9 \rightarrow \text{occupied} \rightarrow 1 \rightarrow 3+5$$

For 1989

$$h_1=9, h_2=6$$

$$i=0 \rightarrow 9$$

$$i=1 \rightarrow (9+6) \bmod 10 = 5$$

$$i=2 \rightarrow (9+6) \bmod 10 = 1$$

$$i=3 \rightarrow (9+6) \bmod 10 = 7$$

b) Open addressing - Linear

$$h_i(x) = h(x+i) \bmod 10$$

Where $i = 0, 1, 2, 3, \dots$

Insert 1 by 1

4371

$h = 1$

$\rightarrow \text{index } 1$

1323

$h = 3, i = 3$

6173 $h = 3 \rightarrow$ occupied

then

$$(3+1) \bmod 10 = 4$$

$i = 4$

4199

$h = 9, i = 9$

4344

$h = 4$

9679

$h = 9 \rightarrow$ occupied

$i = 0$

1989

$h = 9 \rightarrow$ occupied

$i = 2$

Index	value
0	9679
1	4371
2	1989
3	1323
4	6173
5	4344
6	-
7	-
8	-
9	4199

Open Addressing $\rightarrow$ Quadratic Probing

insert 4371

$h=1 \rightarrow$ index 1

1323

$h=3 \rightarrow$ index 3

6173

$h=3 \rightarrow$ occupied

$i=4$

4189

$h=9 \rightarrow$ ~~occupied~~ index 9

4344

$h=4 \rightarrow$ occupied
index 5

9679

$h=9 \rightarrow$ index 0

1989

$h=9 \rightarrow$ occupied - index 11

for $i=2$

$$(a + i^2) \text{ Mod } 10 = (9 + 4) \text{ mod } 10 = 3$$

for $i=3$:

$$(9 + 3^2) = 18 \text{ mod } 10 = 8$$

index 8

Index	Value
0	9679
1	4371
2	-
3	1323
4	6173
5	4344
6	-
7	-
8	1989
9	4199